Section Two: Short answer 35% (70 Marks)

This section has **11** questions. Answer **all** questions. Write your answers in the spaces provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to the appropriate number of significant figures and include appropriate units where applicable.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Suggested working time: 60 minutes.

Question 26 (4 marks)

Write balanced ionic equations for any reactions that occur in the following situations. If no reaction occurs, state **No Reaction**.

(a) Solid copper(II) carbonate is added to dilute hydrochloric acid. (2 marks)

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(b) Barium nitrate solution is added to sodium hydroxide solution. (2 marks)

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Question 27 **(4 marks)**

For each of the following reactions, describe expected **observations**, including any:

- colour changes
- odours
- precipitates (give the colour)
- gases evolved (give the colour or describe as colourless)

(a) Solid sodium hydrogencarbonate is added to dilute hydrochloric acid. (2 marks)

$NaHCO_3(s) + H^+(aq) \rightarrow Na^+(aq) + H_2O(\ell) + CO_2(g)$

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(b) Iron filings are added to silver nitrate solution. (2 marks)

$Fe(s) + 2\ Ag^+(aq) \rightarrow 2\ Ag(s) + Fe^{2+}(aq)$

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Question 28 **(4 marks)**

The table below shows the first four ionisation energies of aluminium.

	1st	2nd	3rd	4th
Ionisation Energies (kJ mol^{-1})	577	1817	2744	11577

Explain why the difference between the 3rd and 4th ionisation energies is greater than the difference between 2nd and 3rd ionisation energies.

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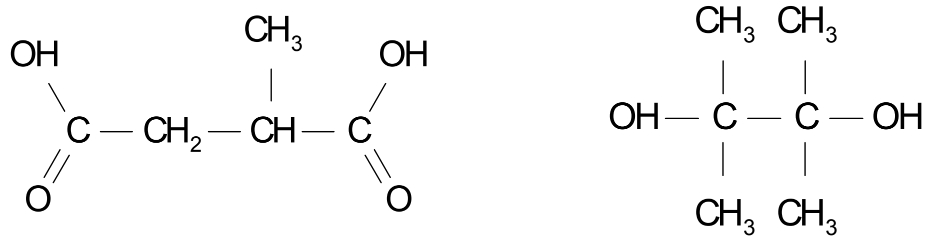
Question 29 **(6 marks)**

A car battery contains sulfuric acid solution, with a concentration of 37.0% by mass. A mechanic accidently spilled 75.0 mL of this solution, which weighed 91.0 g. To treat the spill, 350 mL of 2.00 mol L^{-1} sodium hydroxide solution was added to the spilt acid and 10.0 litres of water added to dilute the resulting solution. Calculate the pH of the final solution, at 25^oC.

Question 30 **(4 marks)**

Consider the two molecules below.



(a) Draw the repeating section of the polymer that would be produced from these two molecules. (3 marks)

(b) Name the other product that is formed during this reaction. (1 mark)

Question 31 **(8 marks)**

A chemist was required to conduct analysis to check that the exact mass of magnesium carbonate present in a 0.500 g indigestion tablet was 460 mg, as claimed by the manufacturer. He decided to carry out the experiment using an indirect (back) titration.

This type of titration involves adding a known amount of acid that is enough to dissolve the tablet and leave an excess, carrying out a titration to calculate the amount of unreacted (excess) acid and using this value to calculate the amount of acid that reacted with the carbonate in the original tablet.

(a) Explain why an indirect (back) titration is the method used. (2 marks)

(b) The chemist had a solution of 1.00 mol L^{-1} hydrochloric acid and a standardised solution of 0.250 mol L^{-1} sodium carbonate to use in the titration. For convenience, he was aiming to have a titre of the sodium carbonate of 20.0 mL. Calculate the volume of the hydrochloric acid he should add to each tablet.

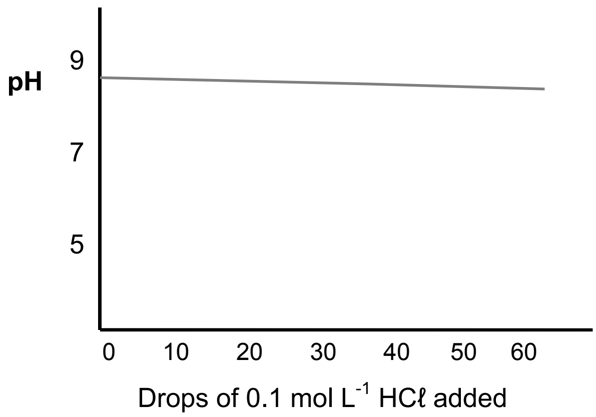
(6 marks)

Question 32 **(6 marks)**

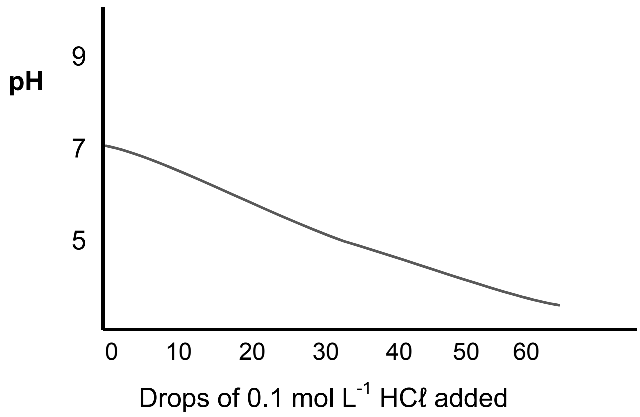
The following two graphs show the effect on the pH of adding dilute hydrochloric acid drop by drop to:

Solution 1: Ammonia (NH_3) solution mixed with ammonium chloride ($NH_4Cℓ$) solution
Solution 2: Sodium chloride ($NaCℓ$) solution.

Graph 1. The change in pH when adding dilute HCℓ to an aqueous mixture of NH_3/$NH_4Cℓ$



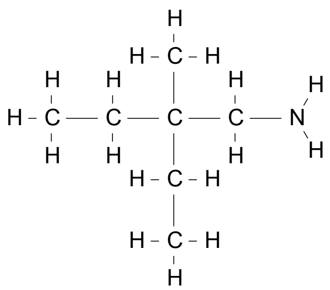
Graph 2. The change in pH when adding dilute HCℓ to a solution of NaCl



Explain, using equations, the differences between the shape of the two graphs.

Question 33 **(3 marks)**

Complete the following table showing the structure and names of three organic compounds.

Structure	IUPAC Name
	methyl propanoate
	hexan-3-one
	

Question 34 **(14 marks)**

(a) For each species listed in the table below, draw the structural formula, representing **all** valence shell electron pairs as **:** or as **—** and indicate the shape of the species by a sketch or a name. (6 marks)

Species	**Electron Dot Diagram (Lewis diagram)**	**Shape**
Carbon monoxide, CO		
Carbon dioxide, CO_2		
hydrogencarbonate ion, HCO_3^-		

(b) Compare the polarities of the carbon dioxide and carbon monoxide molecules, explaining the cause of any differences. (2 marks)

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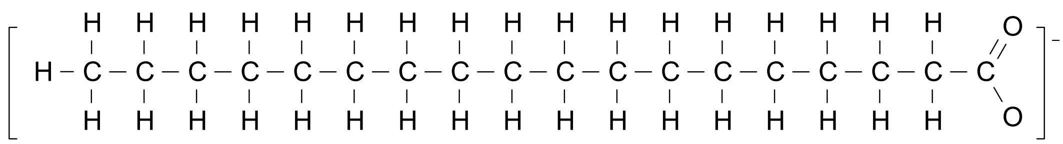
(c) Sodium hydrogencarbonate is soluble in water. Describe, with the aid of a **legible and clearly labelled diagram(s)**, the processes occurring when solid sodium hydrogencarbonate dissolves in water. (3 marks)

(d) Below is the structure of the stearate ion, $CH_3(CH_2)_{16}COO^-$, which is present in a number of types of soap.



Explain why the stearate ion is soluble in water **and** non-polar substances such as oil and grease. (3 marks)

Question 35 **(10 marks)**

Consider the reversible reaction below which is used in the production of quick lime (calcium oxide).

$$CaCO_3(s) + heat \rightleftharpoons CaO(s) + CO_2(g)$$

(a) Write the equilibrium constant expression for this reaction. (1 mark)

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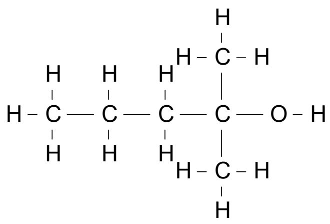
(b) For each of the following changes made to the system at equilibrium, predict the changes to the yield of the reaction once equilibrium is re-established, using the terms increase, decrease or no change. Provide a reason for your prediction in each case, using *reaction rates and collision theory*. (9 marks)

Imposed Change	**Effect on Yield** Write increase, decrease or no change	**Reason for your prediction** (use reaction rates and collision theory)
Increase temperature		
Decrease the volume of the reaction chamber		
Remove calcium oxide as it is produced		

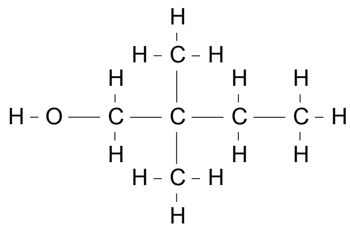
Question 36 **(7 marks)**

Consider the two molecules below.

Molecule A



Molecule B



(a) Write the IUPAC name of the two molecules. (2 marks)

Molecule A

____

Molecule B

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(b) Describe a chemical test that could be used to distinguish between A and B. State the expected observations for both substances. (5 marks)

Substance	Description of chemical test	Expected observations
Molecule A		
Molecule B		

End of Section Two

Section Three: Extended answer 40% (80 Marks)

This section contains **5** questions. You must answer **all** questions. Write your answers in the spaces provided.
Where questions require an explanation and/or description, marks are awarded for the relevant chemical content and also for coherence and clarity of expression. Lists or dot points are unlikely to gain full marks.
Final answers to calculations should be expressed to the appropriate number of significant figures.

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Suggested working time: 70 minutes.

Question 37 **(19 marks)**

This question is about the production of sulfuric acid (H_2SO_4). This process is carried out through a number of steps:

Step 1
Liquid sulfur is reacted with dry air to produce sulfur dioxide (SO_2).

Step 2
The sulfur dioxide is oxidised to sulfur trioxide using vanadium(V) oxide as a catalyst. This step is called the Contact Process. The equation for the reaction is shown below.

$$2\ SO_2(g) + O_2(g) \rightleftharpoons 2\ SO_3(g) \qquad \Delta H = -196\ kJ\ mol^{-1}$$

Step 3
Concentrated sulfuric acid (98.0 % by mass) is used to dissolve sulfur trioxide where it forms oleum ($H_2S_2O_7$).

Step 4
The oleum is mixed with water to obtain more sulfuric acid.

A team of chemical engineers carried out step 2 at a variety of temperatures to inform decisions about the optimum conditions for the reaction. Their results are shown below.

Table 1. Yield of sulphur trioxide for contact process reaction carried out at 150 kPa pressure, with a V_2O_5 catalyst at a range of temperatures.

Temperature of reaction vessel (°C)	**Yield of SO_3 (%)**
200	95
400	92
500	72
600	40
650	31
700	26
800	18
1000	12

(a) On the grid below, display this data with a line graph. (4 marks)

Note: A spare grid is provided at the end of the examination if required

(b) Use your graph to predict the yield of the reaction at 550 °C. (1 mark)

___

(c) Describe the trend shown by these results. (2 marks)

(d) As a result of these findings, the chemical engineer decided to operate the sulfuric acid plant at a temperature of 200 °C. However, the amount of sulfur dioxide produced was very low. Suggest a reason for this. (1 mark)

(e) After further tests, it was decided to operate the plant at 400 °C. With reference to your graph, explain why this temperature, and not a higher temperature, was chosen. (2 marks)

(f) Assuming a yield of 92.0%, Calculate the volume of oxygen, at 400 °C and a pressure of 150 kPa, required to produce 1.00 tonne (1.00×10^6 g) of sulfur trioxide in the Contact Process: (4 marks)

$$2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g)$$

(g) Sulfuric acid is used to produce agricultural fertiliser, including superphosphate, which is a mixture of two calcium salts. The reaction is shown below:

$$Ca_3(PO_4)_2(s) \; + \; 2\,H_2SO_4(aq) \; + \; 4\,H_2O(\ell) \; \rightarrow \; Ca(H_2PO_4)_2(s) \; + \; 2\,CaSO_4{\cdot}2H_2O(s)$$

If 98.0% (by mass) sulfuric acid is used, calculate the mass of the super-phosphate that can be produced from 1.00 tonne of the sulfuric acid. (assume the reaction is 100% efficient) (5 marks)

Question 38 **(12 marks)**

Dopamine is a primary amine that acts as a neurotransmitter, a chemical that sends signals between nerve cells. Levels of dopamine in the brain have been linked to a number of medical conditions, including Parkinson's disease and ADHD. Dopamine contains carbon, nitrogen, hydrogen and oxygen.

Two samples of were analysed to determine its empirical formula.
A 12.1 g sample was combusted in oxygen and produced 27.6 g of carbon dioxide and 7.87 g of water.
A separate 17.2 g sample was found to contain 1.57 g of nitrogen.

(a) Determine the empirical formula of dopamine (7 marks)

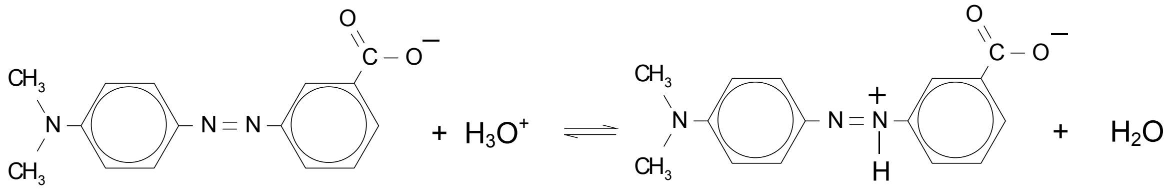
(b) Dopamine is a weak monoprotic base (it can only accept one proton). 10.0 g of dopamine was dissolved in distilled water and the solution made up to 250.0 mL.
25.00 mL of this solution was then titrated against 0.250 mol L^{-1} hydrochloric acid, requiring 26.1 mL of the acid for neutralisation.

From this data, calculate the molecular mass of dopamine, and hence determine the molecular formula of dopamine. (5 marks)

Question 39 **(19 marks)**

Methyl red is an indicator that exists in two different coloured forms, depending on the pH of the solution. In solutions below pH 4.4 the indicator will produce a red colour; above pH 6.2 the indicator appears yellow. Between these pH value's, an orange colouration will appear.

The conversion between the two forms in aqueous solution is shown below.



Yellow form *Red form*

(a) State why the yellow form of methyl red can be described as the conjugate base of the red form of methyl red. (1 mark)

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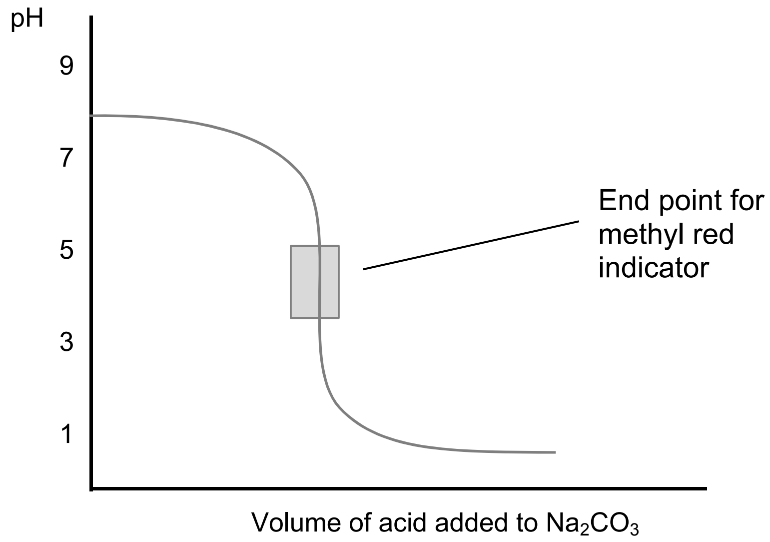
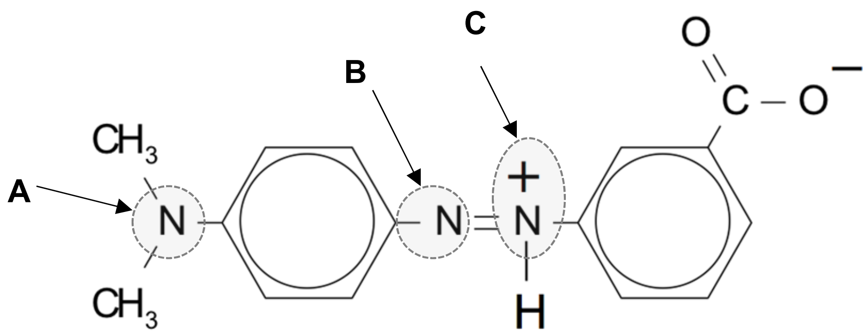
(b) Using your knowledge of equilibrium and reaction rates, use the table below to state and explain how the concentrations of the three ions in the reaction above change when a base such as sodium hydroxide is added. (5 marks)

ION	**CONCENTRATION CHANGE** Write increase, decrease, or no change	**OVERALL EXPLANATION**
Yellow form		
H_3O^+		
Red form		

(c) Explain, using equations and referring to the graph below, why methyl red is a suitable indicator for the titration of hydrochloric acid (burette) with sodium carbonate solution (conical flask). (4 marks)



(d) In the red form of methyl red, the shape of the bonds around each of the three nitrogen atoms **A**, **B** and **C** varies. One is trigonal planar, one is pyramidal and one is bent (v-shaped).



Using the valence shell electron pair repulsion (VSEPR) theory, identify the shape around each of the nitrogen atoms, **A**, **B** and **C** and explain your reasoning, using diagrams where appropriate. (9 marks)

Shape around A	Explanation
Shape around B	**Explanation**

Shape around C	Explanation

Question 40 **(16 marks)**

The Haber-Bosch process involves the reversible reaction of pure hydrogen and nitrogen gas to produce ammonia gas. This reaction is exothermic, releasing 92.4 kJ of heat energy per mole of ammonia. The reaction is conducted under conditions of medium temperature (approximately 500^{o}C), high pressure (approximately 25 500 kPa) and with the use of a catalyst.

(a) Write a balanced thermo-chemical equation for this reaction, including the heat energy as part of the equation. (2 marks)

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(b) Fill in the following table to illustrate how each of the following changes to the system affect the <u>forward reaction rate</u> of this process. (6 marks)

In the Reason column, select the **ONE** reason that <u>best</u> explains the effect.

The abbreviations stand for:
IFC = increased frequency of collisions.
IPS = increased proportion of successful collisions.
LAE = lowering of the activation energy requirement.

CHANGE	EFFECT ON RATE (Write **Increase**, **Decrease**, or **No Change**	REASON Write **IFC**, **IPS**, or **LAE** (see above)
Increase in Temperature		
Decrease in Volume of Reaction Chamber		
Addition of a Catalyst		

(c) One of the methods used to produce the pure hydrogen gas reactant involves reacting methane and steam (H_2O) over a catalyst of NiO to produce carbon monoxide and hydrogen. The carbon monoxide then further reacts with more steam to produce carbon dioxide and hydrogen.

i) Write two separate balanced chemical equations for the two reactions described above. (2 marks)

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ii) How many moles of hydrogen are produced from **ONE** mole of methane? (1 mark)

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(d) One of the major uses of the ammonia product is as a fertiliser. Ammonia is very soluble in water, and can therefore be readily absorbed by plants, enabling them to receive the essential nitrogen needed for growth. This nitrogen cannot be accessed by plants directly from atmospheric gas, so needs to be converted to a soluble form.

i) Write a balanced <u>ionic</u> chemical equation to show the dissolving of ammonia gas in water. (2 marks)

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(e) Compare the boiling point of ammonia to other group 15 hydrides such as PH_3 and AsH_3. State whether it is higher or lower and explain why. (3 marks)

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Question 41 **(14 marks)**

Lawn sand is a mixture of iron compounds (in the form of Fe^{2+}) and sand (mainly SiO_2). It is used to kill moss in lawns. An experiment was carried out to determine the percentage of iron in a sample of lawn sand. The method and student's results are shown below.

Chemicals:

sample of lawn sand
oxalic acid dihydrate ($H_2C_2O_4.2H_2O$) (dried in oven)
1.00 mol L^{-1} sulfuric acid (H_2SO_4)
approximately 0.01 mol L^{-1} potassium permanganate ($KMnO_4$) solution
distilled water

Method Outline	Student's results and notes
1. Make up 250.0 mL of a standard solution of oxalic acid.	mass of oxalic acid dihydrate ($H_2C_2O_4.2H_2O$) dissolved in 250.0 mL**= 0.920 g**
2. Titrate the potassium permanganate solution against standard oxalic acid solution to determine its accurate concentration.	volume of oxalic acid **= 25.00 mL** average volume of $KMnO_4$ **= 26.1 mL**
3. Weigh out the lawn sand and dissolve it in 100 mL of 1.00 molL^{-1} sulfuric acid.	Mass of lawn sand used **= 10.21 g**
4. Filter and make the filtrate up to 500.0 mL with distilled water in a volumetric flask.	Filtrate is pale green solution Residue contains sand and other impurities
5. Titrate a 25.00 mL sample of this solution against the standardised potassium permanganate solution.	(see table below)

	Trials			
	Rough	1	2	3
Final volume (mL)	18.10	35.50	18.55	35.90
Initial volume (mL)	0.00	18.10	1.10	18.55
Titre (mL)	18.1	17.40	17.45	17.35

The relevant half-equations are shown below:

$H_2C_2O_4(aq) \rightarrow 2\ CO_2(g) + 2\ H^+(aq) + 2\ e^-$

$MnO_4^-(aq) + 8\ H^+(aq) + 5\ e^- \rightarrow Mn^{2+}(aq) + + 4\ H_2O(\ell)$

$Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$

In the calculations below, give your final answers to three (3) significant figures.

(a) Calculate the concentration of the oxalic acid solution. (2 marks)

(b) Calculate the concentration of the potassium permanganate solution. (4 marks)

(c) Calculate the total moles of iron (II) in the lawn sand. (5 marks)

(d) Calculate the percentage by mass of iron in the lawn sand. (3 marks)

End of Questions

Spare grid for Question 37

Additional Working Space